Zheng (Bella) CHEN **Phone: +852 55363000 Email: zchenin@connect.ust.hk**

EDUCATION

Hong Kong University of Science and Technology | Hong Kong SAR **Sep 2024 – Dec 2027**
- Ph.D. student in Computer Science and Technology, supervised by Prof. Bo LI
- Research Interest: Mechanistic Interpretability, Representational geometry, Activation steering, and Model reliability.

The University of Hong Kong | Hong Kong SAR **Sep 2020 – Jul 2024**
- Bachelor of Science with a minor in Finance.
- Courses with distinction: Statistical Machine Learning, Linear Statistical Analysis, Natural Language Processing, Image Processing and Computer Vision, Survival Analysis, Time-series Analysis, Optimization, Investments and Portfolio Analysis.

RESEARCH EXPERIENCE

Where the Answer Takes Shape: Shape-Change Peaks Mark Computational Boundaries in LLMs

***Zheng CHEN**, Jiayu An, Futian Jiang, Nitin Jain, Bo Li Under submission* **Under Submission**

- Proposed a label-only shape-change signal that locates, without any intervention, the layers at which an LLM's answer becomes controllable; across six models and ten tasks, 71–83% of causal boundaries fall within one layer of its peaks.

PCA-guided Activation Scaling for Monotonic Bidirectional Control over LLM Sycophancy

***Zheng CHEN**, Zhaoxin Feng, Yip Tin Po, Jianfei Ma, Emmanuele Chersoni, Bo Li* **COLM 2026 (Poster)**

- Proposed a PCA-guided activation scaling method that enables monotonic bidirectional control of LLM sycophancy; demonstrated. that the same steering technique yields substantially different outcomes across models, motivating systematic multi-level evaluation.

Good Arguments Against the People Pleasers: How Reasoning Mitigates (Yet Masks) LLM Sycophancy

*Zhaoxin Feng, **Zheng CHEN**, Jianfei Ma, Yip Tin Po, Emmanuele Chersoni, Bo Li* **ACL 2026 (Poster)**

- Analyzed how CoT reasoning both mitigates and masks LLM sycophancy across objective/subjective tasks, using Tuned Lens on open source models to show sycophancy evolves dynamically during reasoning.

LogiConBench: Benchmarking Logical Consistencies of LLMs

***Zheng CHEN,** Chuan ZHOU, Fengxiang CHENG, Yip Tin Po, …, Bo LI, Zhouchen LIN* **ICLR 2026 (Poster)**

- Designed LogiConBench, a scalable benchmark that generates unlimited, controllable-depth logical reasoning data and remains. challenging for state-of-the-art LLMs, which enables systematic evaluation of their logical consistency.

IPAD: Inverse Prompt for AI Detection-A Robust and Interpretable LLM-Generated Text Detector

***Zheng CHEN**, Yushi FENG, Changyang HE, Yue DENG, Hongxi PU, Bo LI* **NeurIPS 2025 (Poster)**

- Developed IPAD, a robust and interpretable AI-generated text detector that reconstructs the underlying prompt of an input text and verifies its consistency through dual LLM-based distinguishers.

RADI: LLMs as World Models for Robotic Action Decomposition and Imagination

*Dongqi ZUO, **Zheng CHEN**, Chuan Zhou, Yandong GUO, Xiao HE, Mingming GONG* **ICLR 2025 (Workshop)**

- Proposed RADI, a robotic task-planning framework that enables LLMs to act as world models by decomposing complex tasks into. atomic actions and "imagining" environment state changes, which significantly improves planning success rates in VirtualHome.

Can LLMs Recognize Their Own Analogical Hallucinations? Evaluating Uncertainty Estimation for Analogical Reasoning

***Zheng CHEN**, Zhaoxin FENG, Jianfei MA, Jiexi XU, Bo LI* **ACL 2025 (Workshop)**

- Research on black-box uncertainty estimation for analogical reasoning in LLMs, showing that Kernel Language Entropy and Lexical Similarity reliably detect hallucinated analogy transfers across math and code benchmarks.

Meetalk: Retrieval-Augmented and Adaptively Personalized Meeting Summarization with Knowledge Learning from User Corrections

***Zheng CHEN,** Jiang Futian, Yue Deng, Changyang He, Bo Li* **ACL 2025 (Workshop)**

- Designed Meetalk, a retrieval-augmented, feedback-adaptive meeting summarization system that personalizes LLM outputs through user-corrected knowledge bases, beating an iFLYTEK ASR + GPT-4o baseline in a real-world user study.

Persuasion or Insulting? Unpacking Discursive Strategies of Gender Debate in Everyday Feminism in China

*Yue Deng, **Zheng Chen,** Changyang He, Zhicong Lu, Bo Li* **CHI 2024 (Poster)**

- Analyzed uncivil and constructive discourse in Chinese online gender debates, identifying 15 discursive strategies and how they shape user engagement and responses.

INDUSTRY EXPERIENCE

J.P. Morgan Asset Management – Data Science Team Summer Associate | Hong Kong **Jun – Sep 2026**

- Developed and evaluated agentic guardrail mechanisms for LLM-based agents.

LEADERSHIP EXPERIENCE

- Selected for the **Hong Kong Cyberport Micro Fund** with HKD 100,000 for a personalized AI Agent tool. *(Mar–May 2024)*•
  Received HKD 100,000 from the **HKSTP Ideation Program Fund** for a skiing safety device design. *(Jun–Sep 2024)*
- Volunteered in **World Vision Zambia** to improve the borehole system and community sanitation. *(May–Aug 2022)*
- Selected as a **Laidlaw Scholar**, conducted social media analytics research in UCL. *(Feb 2021)*

SKILLS AND INTERESTS

Programming and Algorithms: C++, Python, SQL, PyTorch, Transformers, Diffusion models, SAE lens,

Languages: Mandarin (Native), Cantonese (Native), English (IELTS 8), German (Basic)

Interests/Experiences: Entrepreneurship, Traveling, Snowboarding, Skiing, Scuba Diving, Bouldering